

Asymmetric Information

Hidden type, and hidden action

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Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 22** — *Frontiers of Microeconomics*, the first section, on asymmetric information.

Mankiw states these results. This lecture derives them.

Where we are going

1. **Information, and two ways it can be unequal.**
2. **Counting and averaging** — the one piece of machinery needed.
3. **Hidden type** — the market for used cars.
4. **The threshold** — when the market works, and when it does not.
5. **Warranties** — how the informed side can prove what it knows.
6. **Hidden action** — the principal and the agent.
7. **The two conditions** a contract must satisfy.
8. **Limited liability** — what the problem costs, and executive pay.
9. **What the model leaves out.**

PART 1

Information, and Two Ways It Can Be Unequal

What the standard treatment assumes

When a market is drawn as a supply curve and a demand curve crossing, several things are being assumed at once. One of them is rarely said out loud:

Both sides of the trade know the same things about what is being traded.

That assumption has a name. Information is **symmetric** when both sides know the same things.

The buyer knows the quality of the good. The employer knows how hard the employee is working. The insurer knows the risk being insured.

Each of those is often false, and this lecture works out what follows when it is.

The two cases

	Hidden type	Hidden action
What is hidden	<i>a characteristic</i>	<i>a choice</i>
Who knows it	the seller / the insured	the agent
When	<i>before</i> the deal	<i>after</i> the deal
Fixed or chosen	fixed	chosen
Standard name	adverse selection	moral hazard
Analytical name	hidden type	hidden action

A used car *is* good or bad before anyone bargains over it. Effort is *decided* once the contract is signed.

Naming the two cases

Adverse selection (hidden type). One party knows a *characteristic* of what is being traded and the other cannot observe it. The characteristic is fixed before any bargaining.

Moral hazard (hidden action). One party takes an *action* the other cannot observe. The action is chosen after the contract is signed.

The standard names are the ones in the textbooks. The analytical names say what is actually hidden, and they are the ones that tell you which half of the argument applies.

What the analysis has to deliver

For each case, three questions.

1. **Does it break the market?** Not "might it" — under what exact condition.
2. **What does it cost?** A number, not an adjective.
3. **What can be done?** And why does that work rather than merely sounding sensible.

Parts 3 to 5 answer all three for hidden type. Parts 6 to 8 answer them for hidden action.

Active Learning 1

Classify each. Hidden type, hidden action, or neither. Say what is hidden and from whom.

1. A homeowner takes out fire insurance, then stops replacing the batteries in the smoke alarms.
2. An applicant for health insurance knows about a condition the insurer has not asked about.
3. A restaurant raises its prices and loses some customers.
4. A contractor is paid a fixed sum for a job regardless of how long it takes.

Four minutes. The reason matters more than the label.

PART 2

Counting and Averaging

A lot of a hundred cars

This part contains no economics. It builds the one piece of arithmetic the rest of the lecture needs.

A dealer's lot holds **one hundred cars**. Sixty are high quality and worth \$10,000 to a buyer. Forty are low quality and worth \$4,000.

A buyer picks one without being able to tell which kind it is. **What is it worth?**

Step 1 — what is the whole lot worth?

	how many	each worth	together
High quality	60	\$10,000	\$600,000
Low quality	40	\$4,000	\$160,000
All	100		\$760,000

Sixty cars at ten thousand, forty at four thousand. Nothing here is more than multiplication and addition.

Step 2 — what is one car worth?

$$\frac{\$760,000}{100} = \$7,600$$

The average value of a car on this lot. A buyer who cannot tell the two kinds apart is buying exactly this: one car drawn from this lot.

Note that **no car is worth \$7,600**. Every car is worth either \$10,000 or \$4,000.

Step 3 — the same arithmetic, rearranged

$$\underbrace{\frac{60}{100}}_{\text{share high}} \times \$10,000 + \underbrace{\frac{40}{100}}_{\text{share low}} \times \$4,000 = \$7,600$$

Identical answer. Multiply first and add, or add first and divide — the arithmetic does not care.

The counts have become **shares**, and the shares add to one.

Step 4 — a name and a symbol

Write λ for the share that is high quality. Then $1 - \lambda$ is the rest.

$$\text{average value} = \lambda v_B^H + (1 - \lambda) v_B^L$$

This is called an **expected value**. It is the average of a list, written so that the shares appear instead of the counts.

That is the only sense in which the word “expected” is used in this lecture.

A different lot: **eighty** cars, of which **fifty** are worth \$12,000 to a buyer and **thirty** are worth \$5,000.

1. Find the total value of the lot, then the value of one car, by dividing.
2. Find it again as shares times values. Confirm the two agree.
3. Write down λ and $1 - \lambda$ for this lot.
4. Is any single car worth the answer you computed?

Four minutes. Show the arithmetic both ways.

PART 3

Hidden Type: The Market for Used Cars

Reservation value. The least a seller will accept for a good, or the most a buyer will pay for it. Below the seller's, no sale; above the buyer's, no purchase.

Below the seller's number there is no sale. Above the buyer's number there is no purchase. **Between the two is where a trade can happen.**

This market is described by **four** such numbers: one for each party, on each of the two kinds of car.

The four numbers in this market

	seller will not accept below	buyer will not pay above	gap
High quality	\$8,000	\$10,000	\$2,000
Low quality	\$2,000	\$4,000	\$2,000

A price inside the gap makes both parties better off. Every car here has such a gap.

Read the last column. The gap is **the same size on both rows**. No car in this market is a bad trade.

Every trade is worth doing

There is no bad car in this market.

A low-quality car is worth \$2,000 to its owner and \$4,000 to a buyer. At a price of \$3,000 both gain \$1,000. **It ought to sell.**

A high-quality car is worth \$8,000 to its owner and \$10,000 to a buyer. At \$9,000 both gain \$1,000. **It ought to sell too.**

All hundred trades are worth making. That is what will make the failure a failure.

Private information. Something one party to a trade knows and the other cannot observe, at any cost worth paying.

- **The seller knows the type.** Which kind of car this is, from years of driving it.
- **The buyer cannot observe it.** Not that the buyer has not been told — the buyer cannot find out. A test drive and an hour with the service history do not settle it.
- **The buyer does know the composition of the pool** — what share of cars are of each kind.

That third assumption is doing work. The buyer is not ignorant, and is not being fooled. **The buyer knows everything except which car this is.**

What will a buyer pay?

A buyer facing an unidentified car will not pay more than what such a car is worth on average:

$$P^{\max} = \lambda v_B^H + (1 - \lambda) v_B^L = 10\lambda + 4(1 - \lambda)$$

This is Part 2's arithmetic, unchanged, with the buyer's two reservation values in it.

Now compare $P^{\max}$ with what the owner of a good car will accept.

Using $v_B^H = \$10,000$ and $v_B^L = \$4,000$:

1. Compute $P^{\max}$ when the lot is 50 high and 50 low.
2. Compute $P^{\max}$ when it is 80 high and 20 low.
3. In each case, will the owner of a high-quality car sell? The reservation value is \$8,000.
4. What is happening to $P^{\max}$ as λ rises, and why?

Four minutes. Show the substitution.

PART 4

The Threshold

Step 1 — write the condition

A high-quality car reaches the market only if the most a buyer will pay is at least the least its owner will accept:

$$\underbrace{\lambda v_B^H + (1 - \lambda) v_B^L}_{p^{\max}} \geq \underbrace{v_S^H}_{\text{reservation value}}$$

One inequality. Everything in this part is that inequality, solved and then looked at.

Step 2 — solve it

$$\lambda v_B^H + (1 - \lambda)v_B^L \geq v_S^H$$

$$\lambda v_B^H + v_B^L - \lambda v_B^L \geq v_S^H$$

$$\lambda (v_B^H - v_B^L) \geq v_S^H - v_B^L$$

$$\lambda \geq \frac{v_S^H - v_B^L}{v_B^H - v_B^L}$$

$$\lambda^* = \frac{v_S^H - v_B^L}{v_B^H - v_B^L}$$

Step 3 — put the numbers in

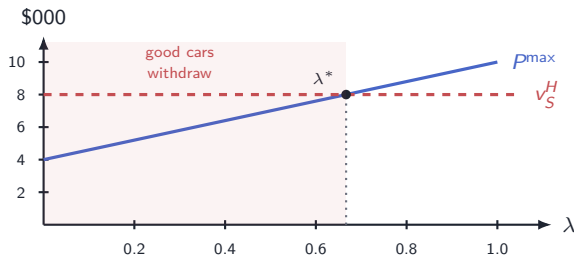
$$\lambda^* = \frac{v_S^H - v_B^L}{v_B^H - v_B^L} = \frac{8 - 4}{10 - 4} = \frac{4}{6} = \frac{2}{3}$$

At least **two thirds** of the lot must be high quality, or the high-quality cars are not offered at all.

Check it directly: at $\lambda = \frac{2}{3}$, $P^{\max} = 10(\frac{2}{3}) + 4(\frac{1}{3}) = \frac{20}{3} + \frac{4}{3} = 8$. ✓

Step 4 — draw it

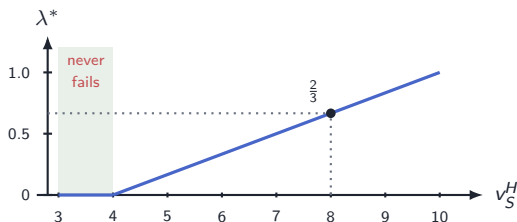
Across: λ , the share of the lot that is high quality. Up: dollars, in thousands.



The rising line is $P^{\max}$, the buyer's ceiling. The flat dashed line is v_S^H , the good owner's floor. **Look at the crossing:** left of λ^* the ceiling is below the floor, and the good cars are not offered.

How the threshold responds

Across: v_S^H , what the owner of a good car will accept. Up: λ^* , the share of good cars the market needs to survive.



- The rising line is λ^* . The marked point is the case in hand: $v_S^H = 8$ gives $\lambda^* = \frac{2}{3}$.
- At the left, $v_B^L \geq v_S^H$: buyers value even a bad car at least as much as a good owner's floor, so the threshold is zero and the market never fails at all.

Three comparative statics

Change one reservation value at a time, leave the others alone, and read the new threshold off the formula.

Change	Threshold	Because
v_S^H rises $8 \rightarrow 9$	$\frac{2}{3} \rightarrow \frac{5}{6}$	good cars are dearer to keep in
v_B^H rises $10 \rightarrow 12$	$\frac{2}{3} \rightarrow \frac{1}{2}$	buyers pay more for the good ones
v_B^L rises $4 \rightarrow 6$	$\frac{2}{3} \rightarrow \frac{1}{2}$	the floor under the pool is higher

v_S^L appears nowhere. What the owner of a bad car thinks it is worth has no bearing on whether the good cars stay.

What it costs

Take a lot that fails the condition: **60 high, 40 low**, so $\lambda = 0.6$ and $P^{\max} = \$7,600 < \$8,000$.

	quality observable	quality hidden
High-quality cars traded	60	0
Low-quality cars traded	40	40
Surplus created	\$200,000	\$80,000

$$60 \times \$2,000 = \$120,000 \text{ of surplus destroyed}$$

Active Learning 4

Second-hand machine tools. Reservation values, in thousands:

	seller	buyer
Good condition	30	36
Poor condition	10	18

1. Compute λ^* from the general expression.
2. If 70% are in good condition, do the good ones trade?
3. The buyer's value of a poor machine rises to 24. What is λ^* now, and what does that mean?

Five minutes. Part three is the case worth thinking about.

PART 5

Warranties: Proving What You Know

The seller has a problem worth solving

The owner of a high-quality car is losing \$2,000 of surplus, and knows exactly why.

Saying “this one is a good car” does nothing. Every seller would say it, so it carries no information.

What is needed is an action that a low-quality seller **would not want to copy.**

A warranty

The seller offers: **if the car needs a major repair within a year, I pay you W .**

	repairs per hundred cars	expected cost of the warranty
High quality	10	$0.10 W$
Low quality	60	$0.60 W$

The same promise costs a low-quality seller **six times** what it costs a high-quality one.

That difference is what makes it a signal rather than talk.

When will each type offer it?

Suppose buyers come to expect that only high-quality cars carry a warranty. A warranted car then sells for $v_B^H = \$10,000$; an unwarranted one for $v_B^L = \$4,000$.

	offers the warranty	does not
High-quality seller	$10 - 0.1W$	4
Low-quality seller	$10 - 0.6W$	4

The warranty must be large enough that a low-quality seller will not copy it, **and small enough** that a high-quality seller still gains from offering it.

The low-quality seller must not want to copy:

$$10 - 0.6W \leq 4 \quad \implies \quad W \geq 10$$

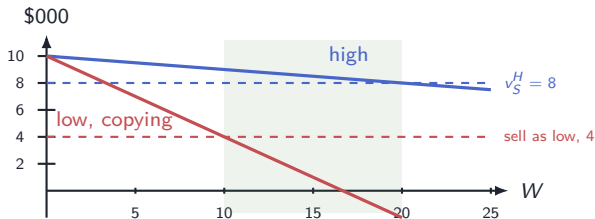
The high-quality seller must still want to sell:

$$10 - 0.1W \geq 8 \quad \implies \quad W \leq 20$$

Any warranty between \$10,000 and \$20,000 separates the two types. Take
 $W = \$12,000$.

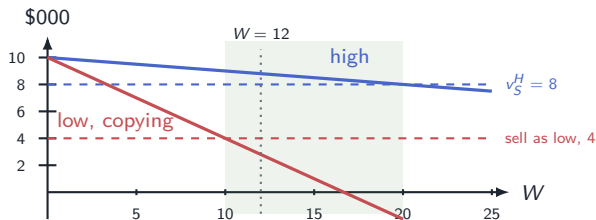
Reading the warranty diagram

Across: the size of the warranty, W . Up: **what the seller ends up with**, in thousands.



- Blue: the high-quality seller's payoff. Red: the low-quality seller *copying* the warranty. Dashed: the alternatives — 8 without a warranty for a good owner, 4 for a bad one.
- The red line is steeper because 60 low-quality cars in a hundred need a repair against 10 high-quality ones. Six times the claims, six times the slope.

Checking a warranty of twelve thousand



- High quality: $10 - 0.1(12) = 8.8 > 8$ — the good seller still beats the floor, so the warranty is offered.
- Low quality copying: $10 - 0.6(12) = 2.8 < 4$ — the bad seller does better selling honestly, so the warranty is not copied.

Both hold, so **twelve separates the types** — which is what the band promised.

What a signal has to be

Signal. A costly action taken to reveal private information, credible only because it is more costly for the party who would falsely take it.

Signal	Cheaper for	Because
Warranty	a reliable good	fewer claims
Money-back offer	a good product	fewer returns
Certification	a competent firm	fewer failures to correct
Probation period	a capable hire	less risk of dismissal

In every case the cost difference does the work. **Sincerity does not enter.**

Active Learning 5

Return to the machine tools: seller values 30 and 10, buyer values 36 and 18, in thousands. Of a hundred good machines, 5 fail in the first year; of a hundred poor ones, 45 do. The seller offers to pay W on failure.

1. Write the condition under which a poor-condition seller will not copy.
2. Write the condition under which a good-condition seller still sells.
3. Give the band, and pick a W inside it.
4. Check your W in both inequalities.

Five minutes. The structure is the same; only the numbers change.

PART 6

Hidden Action: The Principal and the Agent

A different problem

Principal–agent problem. One party (the principal) hires another (the agent) to act on its behalf, and cannot observe what the agent does.

Principal	Agent	Hidden action
Shareholders	the chief executive	managerial effort
An employer	an employee	how hard they work
A client	a lawyer	hours genuinely spent
An insurer	the insured	care taken

What is hidden here is a **choice**, made *after* the contract is signed.

The setup

A board hires a chief executive. Effort is either supplied or not. The board observes the outcome. It does not observe the effort.

	good outcome \$400	bad outcome \$200	average output
Effort supplied	75 in 100	25 in 100	\$350
Effort not supplied	45 in 100	55 in 100	\$290

Effort costs the executive **\$36** in forgone leisure and attention. The next-best job is worth **\$24**. All figures in thousands.

The outcome is an imperfect guide to effort. Read the bottom row: 45 in every 100 who supply no effort still deliver the good outcome. A good year is not proof of effort, and a bad year is not proof of its absence.

Is effort worth having?

Effort raises average output from \$290 to \$350:

$$350 - 290 = \$60$$

It costs the executive \$36.

$\$60 > \36 , so the effort is worth having. Everything that follows is about how to get it.

Note this is settled **before** any contract is written.

Active Learning 6

A restaurant owner hires a manager. Of a hundred managers who supply effort, 60 achieve takings of \$500 and 40 achieve \$300. Of a hundred who do not, 20 achieve \$500. Effort costs the manager \$30. All in thousands.

1. Average takings with effort, and without.
2. Is the effort worth having?
3. Name the principal, the agent and the hidden action.
4. Why can the owner not simply write “supply effort” into the contract?

Five minutes. Show the averages.

PART 7

The Two Conditions a Contract Must Satisfy

What a contract can say

The contract cannot mention effort, because effort cannot be verified. It can mention the outcome, because the outcome can.

w_H if the outcome is \$400, w_L if it is \$200

Two numbers. The board chooses them, and the executive then chooses effort knowing what they are.

The board must choose them so that the executive wants to supply effort.

Condition 1 — incentive compatibility

Incentive compatibility constraint. The requirement that the action the principal wants is the action the agent prefers, given the contract.

$$\underbrace{0.75w_H + 0.25w_L - 36}_{\text{supplies effort}} \geq \underbrace{0.45w_H + 0.55w_L}_{\text{does not}}$$

The board cannot require effort, because it cannot see it. So it must make effort the executive's own preference. **This inequality is that requirement, written down.**

Solving condition 1

$$0.75w_H + 0.25w_L - 36 \geq 0.45w_H + 0.55w_L$$

$$(0.75 - 0.45)w_H - (0.55 - 0.25)w_L \geq 36$$

$$0.30(w_H - w_L) \geq 36$$

$$w_H - w_L \geq 120$$

The levels drop out and only the gap survives. It does not matter what the executive is paid. It matters how much more the good outcome pays than the bad one.

Condition 2 — the executive must be willing to sign

Participation constraint. The requirement that the agent is at least as well off taking the contract as taking the next-best alternative.

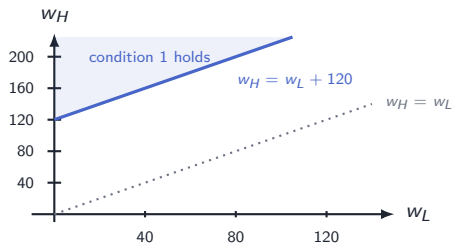
$$\underbrace{0.75w_H + 0.25w_L - 36}_{\text{taking the job, with effort}} \geq \underbrace{24}_{\text{the next-best job}}$$

Two conditions, two different jobs.

Condition 1 compares working hard with *shirking in this job*. **Condition 2** compares this job with *a different job*.

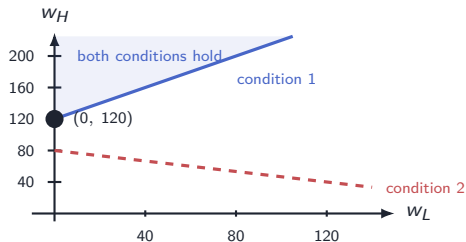
The space of contracts

A contract is a pair of numbers. Across: w_L , the wage in the bad outcome. Up: w_H , the wage in the good outcome. **Every point here is a contract.**



Condition 1 sits **120 above the 45-degree line**, because 120 is the gap it requires. Anything on or above it has a big enough gap.

Where both conditions hold



- Condition 2 slopes the *other* way, because it is about the **average** wage, not the gap.
- The marked point is **the cheapest contract in the feasible region**. It sits on the vertical axis because w_L is pushed as low as it can go — and how low that is comes next.

Active Learning 7

Back to the restaurant. Of a hundred managers with effort, 60 achieve \$500; without, 20 do. The bad outcome is \$300. Effort costs \$30 and the outside option is \$18.

1. Write condition 1 and reduce it to a statement about $w_H - w_L$.
2. Write condition 2.
3. If $w_L = 0$, what is the smallest w_H satisfying condition 1?
4. Does that w_H also satisfy condition 2?

Six minutes. Show the collection of terms in part one.

PART 8

Limited Liability, the Cost, and Executive Pay

First, a case where nothing is lost

Suppose the executive can be made the **residual claimant**: buy the firm for a fixed fee F and keep whatever it earns.

Outcome \$400	the executive keeps $400 - F$
Outcome \$200	the executive keeps $200 - F$
The gap	$400 - 200 = 200$, far above the 120 required

Condition 2 then fixes F : the board's cost is $24 + 36 = \$60$, exactly the outside option plus the cost of effort.

Hidden action costs nothing here.

Limited liability. A wage cannot be negative. The agent can be paid nothing, but cannot be required to pay the principal.

$$w_L \geq 0, \text{ and } w_H \geq 0.$$

The residual-claimant contract required $w_L = -90$. It is unavailable.

So set $w_L = 0$ — the lowest the constraint allows — and see what condition 1 then demands of w_H .

Both conditions with the bad-state wage at zero

Set $w_L = 0$. Limited liability allows nothing lower, and every dollar of w_L would have to be matched in w_H to hold the gap.

Condition 1. The gap is $w_H - 0 = w_H$, so

$$w_H \geq 120$$

Condition 2.

$$0.75w_H - 36 \geq 24 \implies 0.75w_H \geq 60 \implies w_H \geq 80$$

The 0.75 is there because 75 of every 100 executives who supply effort deliver the good outcome — so that is the share of the time w_H is actually paid.

Which one binds

Both must hold, so w_H must clear the **larger** of the two floors. 120 beats 80.

Condition 1 binds; condition 2 has room to spare.

At $w_H = 120$ the executive's expected position is

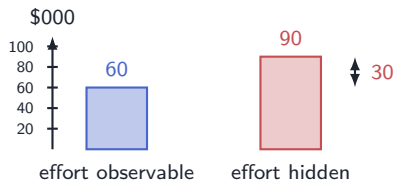
$$0.75(120) - 36 = 90 - 36 = 54$$

against an outside option of 24 — **30 more than the minimum needed to sign**, and the board cannot claw it back. Cutting w_H breaks the incentive condition; raising w_L narrows the gap and breaks the same one.

Expected wage bill: $0.75 \times 120 = \$90$.

What the problem costs

Up: the board's expected cost of getting the effort, in thousands.



60 is what the board would pay if effort were observable: the outside option of 24, plus the cost of effort, 36. **90** is what it pays when effort is hidden: 0.75×120 . The difference, $\$90 - \$60 = \$30$, is the cost of not being able to observe effort.

The 30 goes to the executive, above the outside option. It is not burned.

Executive pay

The contract has a slope:

$$\frac{w_H - w_L}{y_H - y_L} = \frac{120 - 0}{400 - 200} = 0.6$$

The executive bears 60 cents of every extra dollar the firm produces — and the firm's outcome is only partly within the executive's control.

Observed pay–performance sensitivity is far below this. The model is missing something, and Part 9 says what.

Active Learning 8

The restaurant again: gap required 75, outside option 18, effort cost 30, outcomes \$500 and \$300, and 60 in a hundred achieve \$500 with effort.

1. With $w_L = 0$, which condition binds, and what is w_H ?
2. Compute the owner's expected wage bill.
3. Compute the first best, and hence the agency cost.
4. Compute the slope of the contract.

Six minutes. Part one was settled in Active Learning 7.

PART 9

What the Model Leaves Out

Four assumptions doing real work

- **Two types, two outcomes, two effort levels.** Real quality and real effort are continuous. The direction of every result survives; the clean thresholds do not.
- **Nobody minds risk.** The whole of Part 8's transfer becomes a genuine loss once the agent must be paid to bear an outcome they do not control.
- **One period, one transaction.** No reputation, no repeat business, no future in which a name is worth protecting.
- **A fixed set of instruments.** A price and a warranty in the first half; two wages in the second.

Two remedies, one principle

	Signalling	Screening
Who acts	the informed party	the uninformed party
What they do	take a costly action	offer a menu of choices
Works when	the cost differs by type	the choice differs by type
Example	a warranty	insurance deductibles

Part 5 built a signal. Screening is its mirror image, and rests on the same condition: **the two types must find the same option differently attractive.**

Where this leads

- **Insurance.** Both problems at once: the applicant knows the risk (hidden type) and chooses the care taken (hidden action).
- **Credit.** Lenders cannot tell good borrowers from bad, and raising the interest rate worsens the pool — so lending is rationed rather than priced.
- **Labor.** Education as a signal, probation as screening, performance pay against hidden action.
- **Corporate governance.** Boards, auditors and disclosure rules exist because owners cannot observe what managers do.

Each is the same two problems, in a different market.

Active Learning 9

1. A landlord offers two leases: one with a large deposit and low rent, one with no deposit and high rent. Signalling or screening? Who is uninformed?
2. A university degree raises earnings partly because it is harder for less able students to obtain. Which remedy is that, and what condition must hold?
3. In Part 8 the agency cost was a transfer, not a loss. Which assumption made that true?
4. A used-car dealer has traded in the same town for thirty years and offers no warranty. Explain, and say which assumption must be relaxed.

Six minutes. Part four needs a sentence, not a paragraph.

Two problems, told apart by what is hidden

When the hidden thing is a characteristic, the informed side withdraws and the market loses trades that were worth making. When it is an action, the contract must pay for an outcome instead, and the agent collects a surplus that cannot be recovered. In both cases nobody lies, nobody has market power, and the failure is complete.

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